NGUYEN GIA NGHI

Ho Chi Minh City, Vietnam | gianghi30032005@gmail.com | Portfolio | GitHub | LinkedIn

RESEARCH INTERESTS

Software engineering | Distributed and cloud systems | Systems performance and reliability

Debugging, observability and latency diagnosis

EDUCATION

VNUHCM-University of Science Sep 2023–May 2027 (Expected)

Bachelor of Science | Major: Information Technology Ho Chi Minh City

GPA: 8.41/10.00 (3.61/4.00) | 133 credits completed

UNDERGRADUATE RESEARCH THESIS

Research title (English translation): Latency diagnosis and optimization mechanisms for edge cloud-gaming systems

Official title (Vietnamese): Nghiên cứu cơ chế chẩn đoán và tối ưu hóa độ trễ cho hệ thống trò chơi đám mây biên

Supervisor: Professor Le Ngoc Son

Expected completion date: March 2027

- Studying whether latency fingerprints, context-specific patterns of how systems respond to a relief action, can help distinguish cross-layer bottlenecks and support more targeted recovery than generic adaptation across heterogeneous edge-cloud-gaming environments.
- Built a self-hosted edge-cloud-gaming testbed using containerized game workloads, desktop host orchestration, GStreamer capture, and WebRTC streaming. The testbed can also export cross-layer telemetry used for the research.
- Implemented the first end-to-end fingerprinting pipeline which includes versioned records, response-vector construction, fingerprint storage, and fingerprint matching. In an initial repeatability check, a separate experimental run matched the initial seed with a match strength of 0.982 across 22 shared features.
- Planning comparisons with fixed-profile, threshold-based, and passive-telemetry baselines. Comparisons will be evaluated through diagnosis accuracy, P95/P99 latency, frame-deadline misses, recovery time, quality degradation, and probe overhead. A secondary comparison will evaluate stale-work handling with and without deadline awareness.

RELEVANT ACADEMIC PREPARATION

Selected strong results (10-point scale): Discrete Mathematics (9.40) | Introduction to Cloud Computing (9.10) | Cloud Implementations and Operations (9.70) | Computer Systems (9.10) | Software Architecture (8.60) | Software Design (8.50) Applied Statistics for Engineers and Scientists (8.10) | Operating Systems (8.00) | Introduction to Databases (8.00)

SYSTEMS AND SOFTWARE PROJECTS

Latency Fingerprinting | Python research engine | Repository

- Developed a versioned Python pipeline and CLI that can validate records against versioned schemas, ingest testbed data, construct response vectors, and match fingerprints within compatible contexts.

PIXELATED Studio Edition | Creator platform and self-hosted edge-cloud-gaming research testbed

- Built and released a creator platform using TypeScript and React, then extended it into an edge-cloud-gaming testbed that streams interactive workloads from a host owned by users to a lightweight browser client.
- Implemented a Fastify control plane, Electron-based host orchestration, a containerized runtime, GStreamer capture and encoding, and WebRTC streaming, with cross-layer telemetry export.

Repository | Live application | Releases

PIXELATED User Edition | Browser-based WebAssembly gaming platform

- Built and deployed a PWA application that runs NES, Game Boy, and Game Boy Color titles locally in the browser through WebAssembly libretro cores, without relying on the desktop engine or WebRTC streaming.
- Instrumented the browser runtime to export data including launch-time, frame-pacing, long-task, capability, memory, and error telemetry, providing a browser baseline for comparison with the Studio version's edge-cloud-streaming path.

Repository | Live application

INDUSTRY EXPERIENCE

Full-Stack Engineer Intern | TMA Solutions Jun–Aug 2026

Ho Chi Minh City

- Implemented bug fixes, product improvements, and small features for an AI-assisted project-management application built with MongoDB, Express.js, React, and Node.js.
- Modeled and queried MongoDB data through Mongoose, developed and validated API tests, and contributed changes through an Agile/Scrum workflow.

TECHNICAL SKILLS

Programming	TypeScript, JavaScript, Java, Python, C/C++, SQL
Frontend	React, Vite, browser Web APIs
Backend	Node.js, Express.js, Fastify, Spring Boot, REST APIs, Socket.IO
Data	PostgreSQL, MySQL, MongoDB
Cloud platforms	AWS, GCP
Infrastructure and DevOps	Linux, Docker, Kubernetes, Terraform, GitHub Actions
Real-time media	WebRTC, GStreamer
Performance evaluation	Telemetry instrumentation, benchmarking, latency analysis

CERTIFICATIONS AND ENGLISH

- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner
- IELTS Academic: 8.0 overall - Listening: 9.0, Reading: 9.0, Writing: 7.5, Speaking: 7.0 (test date: 2nd October 2025)